

THE ENGINEERING FILAMENT FIELD GUIDE



Engineering filaments are where the X2D earns its keep. Run the model on the main nozzle, the support interface on the auxiliary nozzle, and parts that used to fight you come off clean. This guide maps all eight materials: what each is best at, what to watch for, and which support interface to pair it with.

THE NYLONS

PA6-CF · PA6-GF · PA12-CF · EZ PA

THE ABS FAMILY

ABS-GF · ASA · PC-ABS

THE SPECIALIST

PC

BEFORE YOU PRINT

Six rules that apply to all eight materials.

1

Dry it first, every time

Nylons, PC, and fiber-filled materials drink moisture. Wet filament pops, strings, and prints weak. Dry, then feed from a dry box.

2

Hardened steel for fiber

Anything with CF or GF grinds brass to dust. Use hardened steel and a 0.6 mm nozzle to cut down on clogs.

3

The heated chamber is the point

The X2D's enclosed, actively heated chamber keeps these from warping. Preheat it before you start.

4

Help it stick

A brim plus glue handles all of these. Preheat the plate and lay the first layer down slow.

5

Moderate walls and infill

More material means more shrinkage. Up to 6 walls and up to 50% gyroid infill holds warping down.

6

Ventilate, mask for CF

ABS, ASA, and PC-ABS give off fumes, so vent the room. Sand carbon fiber parts with a mask — the dust is an irritant.

TERMS YOU'LL SEE

HDT — heat deflection temperature: where a part starts softening under load. Higher = handles more heat.

ANNEAL — bake the finished part in an oven to relax stresses and boost strength and heat resistance.

INTERFACE — the support layer touching the model. A plastic the model won't weld to means supports snap away clean.

DRY BOX — a sealed box with desiccant that feeds the printer, keeping filament dry mid-print.

FAMILY 01 · FOUR MATERIALS

THE NYLONS

Strongest, stiffest, thirstiest for moisture. PA6-CF pairs with ASA; the rest with Support for ABS.

PA6-CF

Carbon fiber Nylon 6

HARD TO PRINT

186°C

HDT APPROX

STRENGTHS

- Very high stiffness and strength when dry
- Excellent heat resistance, ~186°C HDT
- Impact, vibration, oil and UV resistant

WATCH FOR

- Soaks up moisture fast — popping, stringing, weak parts
- Warps; carbon fiber is abrasive
- More brittle at extremes than PA6-GF

BEST FOR

Load-bearing structural parts, fixtures, clamps, automotive and mechanical parts in dry conditions.

INTERFACE
ASA FILAMENT

PRINT

Nozzle 280–300°C · Bed 100–120°C · Dry 80°C 8–12h · Hardened 0.6mm · Anneal 80–130°C



SCAN TO
BUY

PA6-GF

Glass fiber Nylon 6

HARD TO PRINT

182°C

HDT APPROX

STRENGTHS

- Exceptionally strong and durable
- More stable, less warp than plain nylon
- Absorbs impact instead of shattering

WATCH FOR

- Very moisture sensitive
- Glass fiber is abrasive; still warps
- Toughness improves slightly damp

BEST FOR

High-impact structural parts, roll cages, brackets — anything surviving repeated shock without cracking.

INTERFACE
SUPPORT FOR ABS

PRINT

Nozzle 280–300°C · Bed 90–100°C · Dry 80°C 8–12h · Hardened 0.6mm · Brim + glue



SCAN TO
BUY

PA12-CF

Carbon fiber Nylon 12

MODERATE

Low

MOISTURE UPTAKE

STRENGTHS

- Low moisture uptake — stable, easier to print than PA6
- Good strength-to-weight
- Chemical resistant

WATCH FOR

- Less stiff, lower heat resistance than PA6-CF; varies by brand
- Still a nylon — drying matters

BEST FOR

Jigs, fixtures, functional and automotive parts where stability beats peak stiffness.

INTERFACE
SUPPORT FOR ABS

PRINT

Nozzle 270–290°C · Enclosure · Dry 80°C ~12h · Hardened 0.6mm · Anneal to improve



SCAN TO
BUY

EZ PA

Sunlu Easy PA — PA6/66 copolymer

EASIER

121°C

HEAT RESISTANCE

STRENGTHS

- Low shrinkage, little warping — engineered to behave
- Strong, tough, abrasion resistant
- UV resistant; wide speed range

WATCH FOR

- Still a nylon, so dry it
- Lower peak performance than the CF and GF nylons
- Runs a notably low bed temp

BEST FOR

A low-stress entry into nylon: living hinges, brackets, jigs, housings, and gears.

INTERFACE
SUPPORT FOR ABS

PRINT

Nozzle 250–270°C (280 fast) · Bed 30–50°C · Dry 70°C 48h



SCAN TO
BUY

FAMILY 02 · THREE MATERIALS

THE ABS FAMILY

02

Reliable, enclosure-loving, a little smelly. All three pair with Support for ABS.

ABS-GF

Glass fiber ABS

HARD TO PRINT

High

HEAT, FOR ABS

STRENGTHS

- Stiffer, stronger, more stable than plain ABS
- Good heat resistance
- Warps less than ABS on larger parts

WATCH FOR

- Moisture sensitive; fumes, so ventilate
- Glass fiber is abrasive — hardened 0.6mm nozzle

BEST FOR

Rigid indoor structural and functional parts that need more backbone than ABS.

INTERFACE
SUPPORT FOR ABS

PRINT

Nozzle 250–270°C · Bed 90–100°C · Enclosure · Dry first · Hardened 0.6mm



SCAN TO
BUY

ASA

Acrylonitrile Styrene Acrylate

MODERATE

UV ✓

YEARS IN SUN

STRENGTHS

- Years of UV and weather without fading or cracking
- Same mechanical profile as ABS; shape to ~100°C
- Warps slightly less than ABS

WATCH FOR

- Needs an enclosure and ventilation
- Bed adhesion can be finicky

BEST FOR

Outdoor parts: signage, mailboxes, garden fixtures, automotive exterior, drone and RC bodies.

INTERFACE
SUPPORT FOR ABS

PRINT

Nozzle 235–260°C · Bed 90–110°C · Enclosure · Fan low



SCAN TO
BUY

PC-ABS

PC and ABS blend

MODERATE

103°C

HDT APPROX

STRENGTHS

- PC heat resistance with ABS printability
- Strong impact resistance
- Easier than most industrial materials; clean surfaces

WATCH FOR

- Still warps and wants an enclosure
- Fumes; dry before printing

BEST FOR

Functional prototypes, tooling, gears, automotive mounts and handles, electronics housings.

INTERFACE

SUPPORT FOR ABS

PRINT

Nozzle 260–280°C · Bed 100–110°C · Enclosure · Dry 70°C 12h



SCAN TO
BUY

FAMILY 03 · ONE MATERIAL

03

The toughest, most heat-resistant common FDM material. Runs on ASA or Support for ABS.

THE SPECIALIST

PC

Polycarbonate

VERY HARD

130°C+

CONTINUOUS USE

STRENGTHS

Toughest, most heat-resistant common FDM material

Survives impacts that shatter ABS

Chemical, UV, water resistant; can be transparent

WATCH FOR

Warps hardest here — chamber must be hot and preheated

Very moisture sensitive

Prints hot enough to wear brass

BEST FOR

Maximum-performance parts: protective casings, structural components, high-heat environments.

INTERFACE

ASA OR SUPPORT FOR ABS

PRINT

Nozzle 270–300°C · Bed 100–110°C smooth PEI · Heated chamber · Dry 80°C 8h



SCAN TO BUY

ALL EIGHT, AT A GLANCE

MATERIAL	HEAT (APPROX HDT)	STANDOUT TRAIT	DIFFICULTY	INTERFACE
PA6-CF	~186°C	Max stiffness & strength	Hard	ASA
PA6-GF	~182°C	Impact — bends, not breaks	Hard	Support for ABS
PA12-CF	Lower, varies	Stable, easier nylon	Moderate	Support for ABS
EZ PA	~121°C	Easiest nylon to print	Easier	Support for ABS
ABS-GF	High for ABS	Rigid indoor structural	Hard	Support for ABS
ASA	~100°C	UV & weather proof	Moderate	Support for ABS
PC-ABS	~103°C	PC heat, ABS ease	Moderate	Support for ABS
PC	~117°C (130°C+ use)	Toughest, most heat	Very hard	ASA / Support for ABS

Heat figures are approximate and vary by brand and test method. Treat them as a ranking, not a spec.

WHEN IT GOES WRONG

SYMPTOM	LIKELY CAUSE	FIX
Popping, steam, fuzzy stringing	Wet filament	Dry it (80°C, 8–12h), then feed from a dry box
Corners lifting off the bed	Chamber or bed too cold	Preheat the chamber , add brim + glue
Weak, chalky layer bonds	Printed wet or too cool	Dry first , raise nozzle temp toward the top of range
Clogs on CF / GF filament	Worn brass nozzle	Hardened steel 0.6mm nozzle
Supports fused to the part	Wrong interface material	Pair per page 5 — run the interface on the aux nozzle

WHICH ONE SHOULD I PRINT IT IN?

It lives outdoors and can't fade or crack

ASA

It has to take the most heat I can throw at it

PC

I want PC-level heat that prints more easily

PC-ABS

I need maximum stiffness and strength

PA6-CF

It has to survive impacts without shattering

PA6-GF

I want nylon strength with the least hassle

EZ PA

It's a rigid indoor structural part

ABS-GF

I want stability and an easier nylon to print

PA12-CF

SUPPORT PAIRING, IN ONE PLACE

Model on the main nozzle, interface on the auxiliary nozzle. The interface is a plastic the model won't weld to — supports snap away instead of fusing on.

PA6-CF



ASA FILAMENT

PA6-GF · PA12-CF · EZ PA · ABS-GF · ASA · PC-ABS



SUPPORT FOR ABS

PC – SURFACE TRADE-OFF



ASA OR SUPPORT FOR ABS

TPU - FLEXIBLE



SUPPORT FOR PLA / PETG



Pairings come from our own test prints on the X2D. Support for ABS releases cleanly across the ABS family and most nylons – only PA6-CF does better against ASA filament. PC works either way: ASA leaves a cleaner interface, Support for ABS releases easier. TPU runs on Support for PLA or PETG.

PLANET 3DP

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